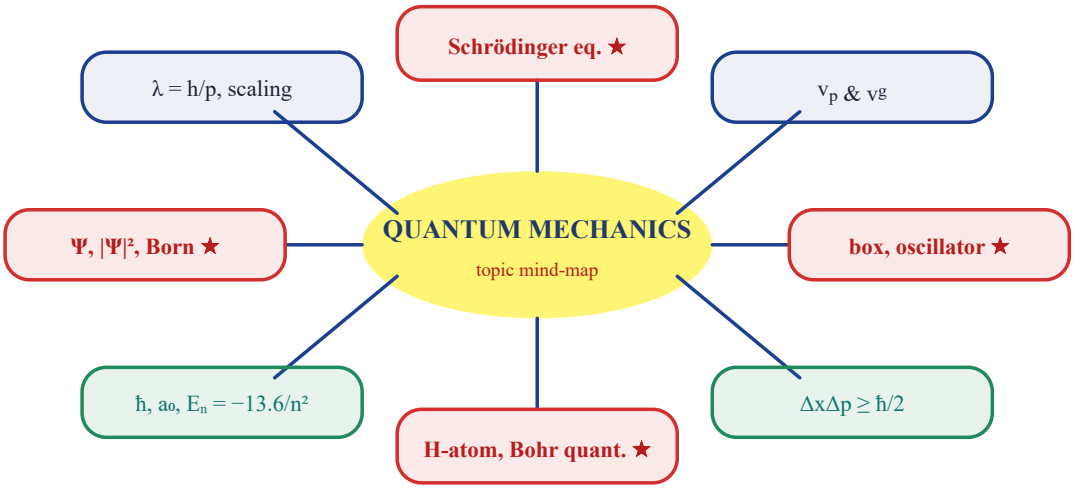
Quantum Mechanics

de Broglie waves · Ψ & Born's rule · Schrödinger equation · box, oscillator, H-atom: exam-ready in 11 sheets



↑ the whole syllabus line on one page: red = high-yield, green = formula-driven

Syllabus coverage

"**De Broglie waves, Phase and group velocity, Wave function and wave equation**." This notebook takes that line to university depth: matter-wave scaling, wave packets, **Born's rule and normalisation**, the Schrödinger equation and separation of variables, the particle in a box, the harmonic oscillator, the hydrogen atom, expectation values, the uncertainty principle and the correspondence principle. Photon momentum $p = h/\lambda$ is in the Quantum Theory of Light sheet.

High-yield list

① $\lambda = h/p = h/mv$
② $v_p = E/p = c^2/v$
③ $v_g = dE/dp = v$
④ $\int|\Psi|^2 dV = 1$, Born rule
⑤ $i\hbar\,\partial\Psi/\partial t = \hat{H}\Psi$
⑥ $E_n = n^2\pi^2\hbar^2/2mL^2$
⑦ $E_n = (n+½)\hbar\omega$
⑧ $E_n = -13.6/n^2$ eV, $L = n\hbar$

Legend: **yellow** = key term · **red circle** = must-memorise value · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

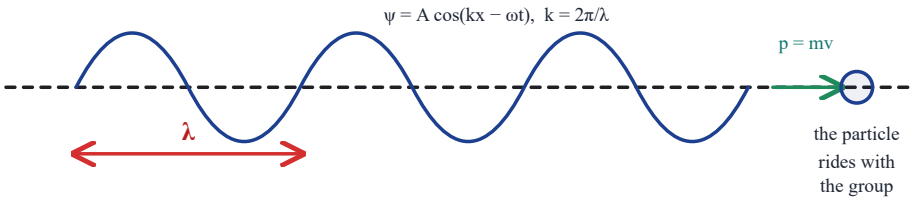
Contents

✓ de Broglie waves in depth …… p. 2
✓ Phase & group velocity …… p. 3
✓ Wave function Ψ & Born's rule …… p. 4
✓ The Schrödinger equation …… p. 5
✓ Particle in a 1-D box …… p. 6
✓ Harmonic oscillator …… p. 7
✓ Hydrogen atom & Bohr …… p. 8
✓ Expectation values & uncertainty … p. 9
✓ Night-before sheet …… p. 10
✓ Practice MCQs + written …… p. 11

1 de Broglie Waves (Matter Waves)

Definition: Louis de Broglie (**1924**, thesis; Nobel Prize **1937**) turned Einstein's logic upside down: if light can behave like particles, then **particles must behave like waves**. Every moving particle of momentum p carries a wave of wavelength λ.

The one relation: **$\lambda = h / p = h / mv = h / \sqrt{(2mK)}$** (non-relativistic)
$h = 6.626 \times 10^{-34}$ J·s (exact since the 2019 SI redefinition) · **$\hbar = h/2\pi = 1.0546 \times 10^{-34}$ J·s**



the wave is attached to the particle; λ is fixed by its momentum

Situation	λ
electron at 1 eV	1.226 nm
electron through 100 V	**0.123 nm = 1.23 Å**
electron through 10 kV	0.123 pm
1 MeV proton or neutron	28.6 fm
cricket ball, 0.1 kg at 10 m/s	**6.6×10^{-34} m**

Memory trick: **"Big thing → tiny wave."** λ ∝ 1/p, so quantum behaviour shows up only for **light or fast** things.

Exam trap: λ = h/p, **not** hp and **not** h/p². λ comes out in metres only if h is in J·s and p in kg·m/s. Above ~0.1c use p = γmv, not mv.

Worked examples ✍️: **(1)** Electron at 2.0×10^{6} m/s: p = $(9.11 \times 10^{-31})(2.0 \times 10^{6}) = 1.82 \times 10^{-24}$ kg·m/s, so λ = $6.626 \times 10^{-34}/1.82 \times 10^{-24}$ = **3.64×10^{-10} m = 0.364 nm**.
(2) 1.0 MeV proton: p = √(2mE) = 2.32×10^{-20} kg·m/s, so λ = **2.86×10^{-14} m = 28.6 fm** — a nucleus-sized wavelength.

Cross-reference: a photon also carries p = h/λ, but with zero rest mass so p = E/c. Davisson–Germer's confirmation (1927) and the Bohr condition 2πr = nλ sit on the Basic Physics de Broglie sheet.

$\lambda_C = h/mc$: the λ of a particle at rest

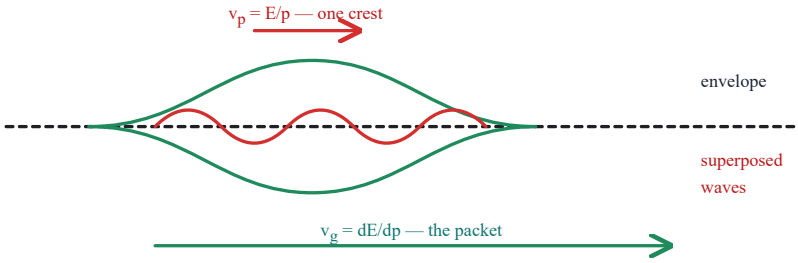
2 Phase Velocity & Group Velocity

Definition: A free particle is not an endless wave — it is a **wave packet**: a superposition of many component waves of nearly equal k. Two speeds describe it, and only one of them is physical.

The two speeds:

Phase velocity $v_p = \omega/k = \mathbf{E/p}$ — the speed of one single crest

Group velocity $v_g = d\omega/dk = \mathbf{dE/dp = v}$ — the speed of the envelope, i.e. of the particle



only the envelope is real: it carries energy and information

	Phase velocity v_p	Group velocity v_g
definition	$\omega/k = E/p$	$d\omega/dk =$ **dE/dp**
free massive particle	**c^2/v ($\geq c$)**	**= v (the particle)**
light in vacuum	c	c
dispersive medium	ω/k	signal & information speed
carries	**no information**	energy **and** information

Derivation: $v_p \cdot v_g = c^2$ ✍️: Free particle: $E^2 = p^2c^2 + m^2c^4$, so $E = pc^2/v$. ① $v_p = E/p$ · ② $v_g = p^2c^2/E$ · ③ product $= c^2$. For light m = 0, E = pc, so both equal c.

Memory trick: "One crest is a fiction, the packet is the truth." A lone crest carries no signal, so its speed is unmeasurable — and may exceed c.

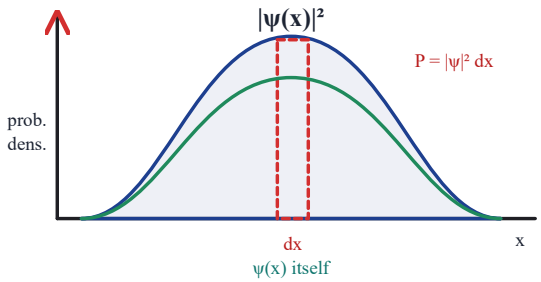
Exam trap: $v_p > c$ breaks nothing. v_g is **dE/dp**, never dp/dm.

Worked example ✍️: Electron at 0.8c: $v_g = 0.8c = 2.40 \times 10^8$ m/s · $v_p = c^2/(0.8c) = 1.25c = 3.75 \times 10^8$ m/s · product $0.8 \times 1.25 = 1$, i.e. c^2 ✓

$p = \gamma mv = 3.64 \times 10^{-22}$ kg·m/s → $\lambda = 1.82$ pm; using mv would have wrongly given 3.03 pm.

3 The Wave Function Ψ & Born's Rule

Definition: **Ψ(x, y, z, t)** is the complete mathematical description of a quantum state: it holds **everything knowable** about the particle. Ψ is **complex-valued** and is **not itself measurable**. **Born's rule** (1926) connects it to the laboratory: $|\Psi|^2$ is the **probability density**.



$|\psi|^2$ is a density; $|\psi|^2 dV$ is a probability — not the same

Object	Meaning
$\Psi(x, t)$	wave function — complex, unobservable
$\|\Psi\|^2 = \Psi^*\Psi$	**probability density** (per unit volume)
$\int\|\Psi\|^2 dV$	**= 1** (normalisation — exactly one particle)
$\|\Psi(x, t)\|^2 dV$	probability of finding it in the element dV
ψ, E	eigenfunction and eigenvalue of $\hat{H}$

- ✓ Single-valued, finite, continuous
- ✓ Ψ and $\partial\Psi/\partial x$ continuous wherever V is finite
- ✓ Ψ obeys superposition (it is linear)
- ✓ $\Psi \equiv 0$ is **not** a state (cannot be normalised)
- ✓ A measurement leaves Ψ in an eigenstate of that quantity
- ✓ Two states add: $\Psi = c_1\Psi_1 + c_2\Psi_2$

Memory trick: "Bar the star, then square": $|\Psi|^2 = \Psi^* \times \Psi$ — multiply by the complex conjugate, never $\Psi \times \Psi$.

Exam trap: $|\Psi|^2$ is a **density**; a probability needs an integral. Ψ may be complex, so Ψ^2 can be complex and meaningless.

Worked example: normalisation ✍️: $\psi = A \sin(\pi x/L)$ for $0 < x < L$.

1. $1 = \int_0^L |A|^2 \sin^2(\pi x/L) dx = A^2(L/2)$ $\int_0^L \sin^2 = L/2$
2. $A = \sqrt{\mathbf{(2/L)}}$, so $\psi_n = \sqrt{(2/L)} \sin(n\pi x/L)$, n = 1, 2, 3 …
3. Probability in the middle half ($L/4 < x < 3L/4$): $P = (2/L)\int \sin^2 dx =$ **1/2**

4 The Schrödinger Equation

Definition: The Schrödinger equation (**1926**; Nobel Prize **1933**) is the **fundamental law of quantum mechanics**, playing the role Newton's second law plays classically. Given Ψ at one instant it fixes Ψ at every later instant. It does **not** say where the particle is — only how the **state** changes.

Both forms:

Time-dependent (always valid): $i\hbar\,\partial\Psi/\partial t = -(\hbar^2/2m)\nabla^2\Psi + V\Psi = \hat{H}\Psi$

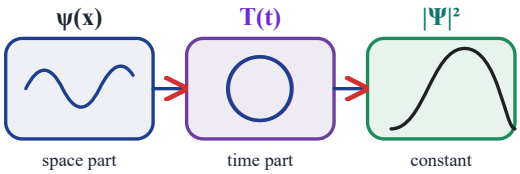
Time-independent (stationary states, V time-independent): $-(\hbar^2/2m)\nabla^2\psi + V\psi = E\psi$

In 1-D just replace ∇^2 by $\partial^2/\partial x^2$.

Classically we write…	…in quantum mechanics it becomes the operator…
position x · momentum p	$x \to x \cdot\ p \to -i\hbar\nabla$
kinetic energy $p^2/2m$	$-(\hbar^2/2m)\nabla^2$
energy E	$E \to i\hbar\,\partial/\partial t$
total energy H	$\hat{H} = -(\hbar^2/2m)\nabla^2 + V$ (Hamiltonian operator)

Separation of variables

1. Put $\Psi(x, t) = \psi(x)\,T(t)$, substitute, divide through by ψT
2. Left side depends on **t**, right side on **position** ⇒ both equal one constant, −E/ħ
3. $T(t) = e^{-iEt/\hbar}$; since |T| = 1, $|\Psi|^2 = |\psi|^2$ is **constant in time**
4. That is a **stationary state**; ψ is the eigenfunction, E the eigenvalue



Ψ = ψ·T, but |T| = 1, so only ψ shapes probabilities

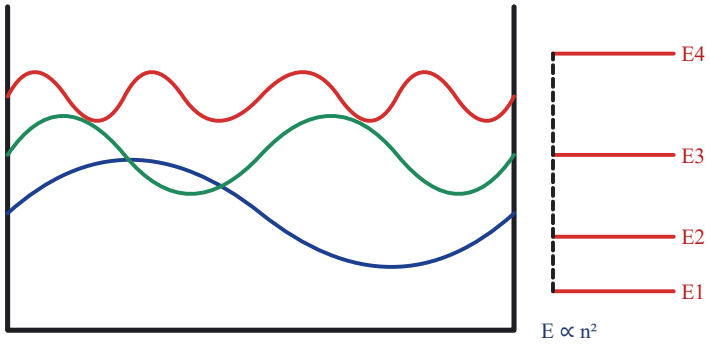
Memory trick: "Replace the word with a hat." Every quantity becomes an **operator** — $p \to -i\hbar\nabla$, $E \to i\hbar\,\partial/\partial t$, $H \to \hat{H}$. Then the equation only says **energy = $\hat{H}$ acting on the state**.

Exam trap: The time-independent form is **invalid** if V depends on time, and it uses **lower-case ψ**. A "stationary state" is stationary in probability only.

Worked example: free particle ✍️: V = 0 gives $-(\hbar^2/2m)d^2\psi/dx^2 = E\psi$. Try $\psi = Ae^{ikx}$, so $d^2\psi/dx^2 = -k^2\psi$: $-(\hbar^2/2m)(-k^2)\psi = E\psi$ ⇒ $\mathbf{E = \hbar^2k^2/2m = h^2/(2m\lambda^2) = p^2/2m}$ — the Schrödinger equation hands back $\lambda = h/p$ for a free particle.

5 Particle in a One-Dimensional Box

The model: $V(x) = \infty$ outside and $V = 0$ for $0 < x < L$ — infinitely hard walls. Because Ψ must stay finite and $V = \infty$ makes the particle impossible there, $\boldsymbol{\psi(0) = \psi(L) = 0}$. Solving $d^2\psi/dx^2 = -(2mE/\hbar^2)\psi$ with those two boundary conditions forces the sine solution to fit a whole number of half-wavelengths.



blue ψ_1 (n = 1), green ψ_2 (n = 2), red ψ_3 (n = 3); the ladder at the right rises as $E \propto n^2$

Quantity	Result
normalised eigenfunction	$\psi_n(x) = \sqrt{(2/L)}\sin(n\pi x/L)$
energy eigenvalue	$\boldsymbol{E_n = n^2\pi^2\hbar^2/2mL^2 = n^2h^2/8mL^2}$
allowed n	n = 1, 2, 3 … (never 0); nodes = n − 1
levels in units of E_1	1 : 4 : 9 : 16 …
momentum	$\langle p\rangle = 0$, but $\langle p^2\rangle = (n\pi\hbar/L)^2$

Memory trick: A particle in a box is **a string fixed at both ends**. Only 3λ/2, 5λ/2, 7λ/2 … fit; λ, 2λ, 4λ never do. Hence n starts at 1 and $E \propto n^2$.

Exam trap: E_1 is **not** zero — it is the **zero-point energy**. And $\langle p\rangle = 0$ does **not** mean $p = 0$; Δp is large (sheet 9).

Worked example: electron in a 0.1 nm box ✍️: $E_1 = h^2/(8mL^2) = (6.626 \times 10^{-34})^2/(8 \times 9.11 \times 10^{-31} \times 10^{-20}) =$ **6.02×10^{-18} J = 37.6 eV**

$E_2 = 150$ eV · $E_3 = 338$ eV · $E_4 = 602$ eV. A 2 → 1 jump releases $3E_1 = 113$ eV, so the emitted photon has $\lambda = 1240/113 =$ **11.0 nm** (far UV).

Halving L lifts every level fourfold

IT link: the barrier between two boxes is a **quantum tunnel** — the electron leaks through even when E < V. That is how NAND flash cells are programmed and erased, and how a scanning tunnelling microscope resolves single atoms (tunnelling current $\propto e^{-2\kappa d}$, Binnig & Rohrer, Nobel 1986).

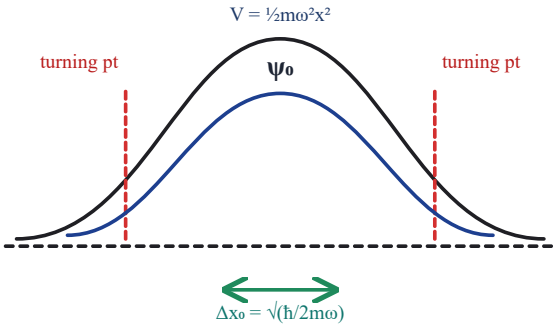
6 The Quantum Harmonic Oscillator

Definition: The 1-D bound system with a quadratic potential: $V(x) = \tfrac{1}{2}kx^2 = \tfrac{1}{2}m\omega^2x^2$, where $\omega = \sqrt{(k/m)}$. It is the most exactly solvable system in quantum mechanics, and the model for molecular vibrations, lattice phonons and LC circuits.

Results:

$\mathbf{E_n = (n + \tfrac{1}{2})\hbar\omega = (n + \tfrac{1}{2})hf}$, $n = 0, 1, 2, 3 \ldots$

$\psi_0 = (m\omega/\pi\hbar)^{1/4}e^{-m\omega x^2/2\hbar}$ · $\langle x\rangle_0 = 0$ · $\langle x^2\rangle_0 = \hbar/(2m\omega)$



Gaussian ground state, narrower than the classical turning points; $E_0 = \tfrac{1}{2}\hbar\omega$ never vanishes

	Classical oscillator	Quantum oscillator
energies	continuous, $E = \tfrac{1}{2}kA^2$	discrete, $(n + \tfrac{1}{2})\hbar\omega$
lowest energy	may be 0	**$\tfrac{1}{2}\hbar\omega$, never 0**
spacing	depends on A	constant, $\hbar\omega$
width	fixed amplitude A	$\Delta x_0 = \sqrt{(\hbar/2m\omega)}$
degeneracy (1-D)	—	none: every level is unique

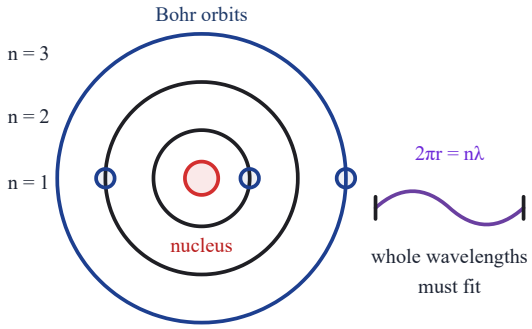
Memory trick: **"The box counts from 1, the oscillator counts from 0."** $n = 0$ would give the forbidden $\psi \equiv 0$ in the box; in the oscillator $n = 0$ is the real ground state with energy $\tfrac{1}{2}\hbar\omega$.

Exam trap: The oscillator is never **dead still**: $\langle p\rangle = 0$ but $\langle p^2\rangle \neq 0$. Its spacing does **not** grow with n, unlike the box.

Worked example ✍️: A molecular vibration with $\omega = 5.0 \times 10^{14}$ rad/s: $\hbar\omega = (1.0546 \times 10^{-34})(5.0 \times 10^{14}) = 5.27 \times 10^{-20}$ J = 0.329 eV, so $E_0 =$ **0.165 eV** and $E_1 - E_0 = 0.329$ eV lands in the infrared. For an electron in the same field $\Delta x_0 = \sqrt{(\hbar/2m\omega)} = 0.34$ nm — ten atomic diameters.

7 The Hydrogen Atom & Bohr Quantisation

Bohr's model (1913): Bohr kept Rutherford's nucleus but forced Planck's quantum on it. **(1)** The electron may move only in orbits whose angular momentum is a whole multiple of ħ — **$L = m_e v\, r = n\hbar$**. **(2)** In these **stationary orbits** it does not radiate; light appears only in jumps between orbits.



an orbit is a standing wave of de Broglie waves

n	1	2	3	4
r_n/Å	0.529	2.12	4.76	8.46
E_n/eV	**−13.6**	−3.40	−1.51	−0.85
L/ħ	1	2	3	4

$a_0 = 4\pi\varepsilon_0\hbar^2/(m_e e^2) =$ **0.529 Å** · $\alpha = 1/137$ · $v_1 = \alpha c = 0.007c$.

Spectral series	n′	Region	Famous lines
Lyman	1	ultraviolet	series limit 91.2 nm
Balmer	2	**visible**	Hα 656.3, Hβ 486.1, Hγ 434.0, Hδ 410.2 nm
Paschen / Brackett	3 / 4	infrared	used in IR spectroscopy of hydrogen

Memory trick: **"Two conditions, one spectrum."** The orbit must hold a whole number of de Broglie wavelengths **and** have L = nħ; balancing the electrostatic pull against the centripetal need then forces $r_n = n^2 a_0$.

Exam trap: **L = nħ is Bohr's, not Schrödinger's.** Real QM gives $L^2 = l(l+1)\hbar^2$ with l = 0 … n−1, so the ground state has L = 0 — a circular orbit is impossible.

Worked example: the Hα line ✍️: Balmer α is the transition n = 3 → n = 2:
$\Delta E = 13.6(1/2^2 - 1/3^2) = 13.6 \times 5/36 =$ **1.89 eV** → $\lambda = 1240/1.89 =$ **656 nm**, hydrogen's red line. Ionising the ground state needs exactly 13.6 eV.

Schrödinger's hydrogen: The energies are the same, $E_n = -13.6/n^2$ eV, but the orbital quantum number l = 0 … n−1 and $m_l = -l \ldots +l$, so the n-th shell holds **n^2 states**: 1s · 2s, 2p · 3s, 3p, 3d …

$1/\lambda = R(1/n_1^2 - 1/n_2^2)$

8 Expectation Values, Uncertainty & Correspondence

Expectation value: An observable A has no definite value in a state Ψ; it has only an average over many identically prepared copies: $\langle A\rangle = \int \Psi^* \hat{A} \Psi \, d\tau$. For $\Psi = \Sigma c_i \Psi_i$ in an orthonormal set, $\langle A\rangle = \Sigma |c_i|^2 \langle A_i\rangle_i$.

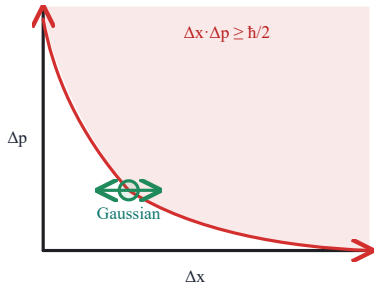
Quantity	Expectation value	Spread (standard deviation)
position	$\langle x\rangle = \int \Psi^* x \Psi \, d\tau$	$\Delta x = \sqrt{(\langle x^2\rangle - \langle x\rangle^2)}$
momentum	$\langle p\rangle = \int \Psi^* (-i\hbar\nabla) \Psi \, d\tau$	$\Delta p = \sqrt{(\langle p^2\rangle - \langle p\rangle^2)}$
energy	$\langle E\rangle = \int \Psi^* \hat{H} \Psi \, d\tau$	ΔE (0 in a stationary state)

Heisenberg uncertainty principle (1927): $\Delta x \cdot \Delta p \geq \hbar/2$

Equality only for a minimum-uncertainty **Gaussian** packet. $\hat{x}$ and $\hat{p}$ do not commute ($[\hat{x}, \hat{p}] = i\hbar$), so no state can be an eigenstate of both.

Correspondence principle (Bohr, 1920)

System	Quantum	Large-n limit
box	$E_n = n^2h^2/8mL^2$	$\Delta E/E = (2n+1)/n^2 \to 0$
oscillator	$E_n = (n+½)\hbar\omega$	constant $\hbar\omega$ → smooth
Bohr H atom	$L = n\hbar$, $r = n^2a_0$	$L/\hbar \to \infty$, orbit looks classical



the product has a floor: tight position blurs momentum

Memory trick: "You may have it small **or** small — never both." Δx is a length, Δp a momentum; their product is an **action** (J·s). Put ħ = 0 and you get the classical world back.

Exam trap: The sign is ≥, and equality is the exception. For the box ground state $\Delta x \Delta p \approx 0.57\ \hbar$, not ħ/2.

Worked example: electron in the 0.1 nm box ✍️: $\langle x\rangle = L/2$ and $\langle x^2\rangle = L^2(1/3 - 1/2\pi^2) = 0.2827\ L^2$, so $\Delta x = \sqrt{(0.2827 - 0.25)}L = 0.181L = 1.81 \times 10^{-11}$ m. Then $\Delta p \geq \hbar/(2\Delta x) = 1.0546 \times 10^{-34}/(3.62 \times 10^{-11}) =$ **2.9×10^{-24} kg·m/s**, i.e. $\Delta v \geq 3.2 \times 10^6$ m/s — while the particle's own speed is only $\sqrt{(2E_1/m)} = 3.6 \times 10^6$ m/s. Position and momentum cannot both be pinned down.

Night-Before-the-Exam Sheet

Constants — memorise:

$h = 6.626 \times 10^{-34}$ J·s

$\hbar = 1.0546 \times 10^{-34}$ J·s

$\hbar/2 = 5.27 \times 10^{-35}$ J·s

$a_0 = 0.529 \times 10^{-10}$ m

$E_1(H) = -13.6$ eV

$\alpha = 1/137$, $v_1 = 2.19 \times 10^6$ m/s

$m_e = 9.11 \times 10^{-31}$ kg

1 eV = 1.602×10^{-19} J

$hc = 1240$ eV·nm

e^- through V: $\lambda(\text{nm}) = 1.226/\sqrt{V}$

Formula sheet:

$\lambda = h/p = h/mv = h/\sqrt{(2mK)}$

$\lambda \propto 1/p$

$v_p = \omega/k = E/p = c^2/v$

$v_g = d\omega/dk = dE/dp = v$

$v_p \cdot v_g = c^2$

light in vacuum: both = c

$|\Psi|^2$ = probability density

$\int|\Psi|^2 d\tau = 1$

$i\hbar\,\partial\Psi/\partial t = \hat{H}\Psi$

$\hat{H} = -(\hbar^2/2m)\nabla^2 + V$

$-(\hbar^2/2m)\nabla^2\psi + V\psi = E\psi$

$\Psi = \psi\, e^{-iEt/\hbar}$

$p \to -i\hbar\nabla$, $E \to i\hbar\,\partial/\partial t$

$\langle A\rangle = \int\Psi^*\,\hat{A}\,\Psi\,d\tau$

$\psi_n = \sqrt{(2/L)}\sin(n\pi x/L)$

$E_n = n^2\pi^2\hbar^2/2mL^2$

$E_n^{osc} = (n+½)\hbar\omega$

$E_0 = ½\hbar\omega$

$r_n = n^2a_0$, $E_n = -13.6/n^2$ eV

$L = n\hbar$, $L^2 = l(l+1)\hbar^2$

$\Delta x \cdot \Delta p \geq \hbar/2$

Top 12 one-liners for MCQs

1. de Broglie: 1924, Nobel 1937
2. $\lambda \propto 1/p$ — double p, halve λ
3. $v_p = c^2/v$, never physical
4. $v_g = dE/dp = v$ (not dp/dm)
5. Born rule: $|\Psi|^2$ is a density
6. Normalisation: $\int|\Psi|^2 d\tau = 1$
7. Schrödinger 1926, Nobel 1933
8. TISE only when V is time-independent
9. Box: n = 1, 2, 3 … and $E \propto n^2$
10. Oscillator: n = 0, 1, 2 … , spacing $\hbar\omega$
11. $a_0 = 0.529$ Å, $E_1(H) = -13.6$ eV
12. Balmer is the visible series

Top traps: $\lambda = h/p$, not hp · $v_p > c$ breaks nothing · $v_g = dE/dp$ · $|\Psi|^2$ is a density · TISE needs V time-independent · box starts at n = 1, oscillator at n = 0 · E_1 and E_0 are never zero · $L = n\hbar$ is Bohr's · $\Delta x\Delta p$ is $\geq \hbar/2$

All tricks together: Big thing → tiny wave · one crest is a fiction, the packet is the truth · bar the star, then square · the box counts from 1, the oscillator from 0 · two conditions, one spectrum · small **or** small, never both

✓ Practice MCQs (exam pattern, 10 marks)

1. de Broglie λ at 100 V:
(a) 1.23 Å (b) 12.3 Å
(c) 0.123 Å (d) 123 Å

2. Time-independent eqn needs:
(a) V varies in time (b) relativistic Ψ
(c) static potential (d) complex Ψ

3. Group velocity, free particle:
(a) ω/k (b) the speed v
(c) c^2/v (d) the speed c

4. $-(\hbar^2/2m)\nabla^2 + V$ is the:
(a) Laplacian (b) wave operator
(c) Lagrangian (d) Hamiltonian

5. Box ground state, width L:
(a) 0 (b) $\hbar^2/2mL^2$
(c) $\pi^2\hbar^2/2mL^2$ (d) $2\pi^2\hbar^2/2mL^2$

6. Oscillator zero-point energy:
(a) 0 (b) ½ħω
(c) ħω (d) 3ħω/2

7. Hydrogen ground state:
(a) −13.6 eV (b) +13.6 eV
(c) −3.40 eV (d) 0 eV

8. Visible hydrogen series:
(a) Lyman (b) Paschen
(c) Brackett (d) Balmer

9. Heisenberg: Δx·Δp?
(a) ΔxΔp = ħ (b) Δx+Δp ≥ ħ
(c) ΔxΔp ≥ ħ/2 (d) ΔxΔp ≤ ħ

10. Quantum → classical when:
(a) ħ → ∞ (b) n → ∞
(c) v → c (d) λ → 0

Answer key → 1-a · 2-c · 3-b · 4-d · 5-c · 6-b · 7-a · 8-d · 9-c · 10-b

Written question: An electron is confined to a 1-D infinite well of width 0.5 nm. (i) Write the time-independent Schrödinger equation inside the well and the boundary conditions. (ii) Obtain the normalised wave function and the energy levels. (iii) Find E_1 in eV. (iv) Find the photon wavelength for n = 3 → n = 1. 8 marks

1. **(i)** Inside the well V = 0, so $-(\hbar^2/2m)d^2\psi/dx^2 = E\psi$ with $\psi(0) = \psi(L) = 0$. 2
2. **(ii)** $\psi = A \sin kx$; $\psi(L) = 0$ forces $k = n\pi/L$, n = 1, 2, 3 … . Normalising, $A^2L/2 = 1 \Rightarrow \psi_n = \sqrt{(2/L)}\sin(n\pi x/L)$, $E_n = n^2h^2/8mL^2$. 2
3. **(iii)** $E_1 = (6.626 \times 10^{-34})^2/(8 \times 9.11 \times 10^{-31} \times 2.5 \times 10^{-19}) = 2.41 \times 10^{-19}$ J = **1.50 eV** 2
4. **(iv)** $\Delta E = (9 - 1)E_1 = 12.0$ eV, so $\lambda = 1240/12.0 =$ **103 nm**, far ultraviolet. 2

Before the exam ✓

- ✓ Convert λ = h/p into nm, Å, fm
- ✓ Explain why $v_p > c$ is fine
- ✓ Separate variables: Ψ = ψT

Also practise:
① Show $v_p v_g = c^2$ from $E^2 = p^2c^2 + m^2c^4$
② Derive $r_n = n^2a_0$ (Bohr)